

MCKV INSTITUTE OF ENGINEERING

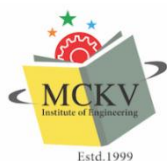
NAAC Accredited "A" Grade Autonomous Institute under UGC Act 1956
 Approved by AICTE & affiliated to Maulana Abul Kalam Azad University of Technology, West Bengal
243 G.T. Road (N), Liluah, Howrah- 711204, West Bengal, India
 Ph: +91 33 26549315/17 Fax +91 33 26549318 Web: www.mckvie.edu.in/

Course Name:	Vehicle Dynamics		
Course Code:	PC-AUE701	Category:	Professional Core Courses
Semester:	Seventh	Credit:	03
L-T-P:	3-0-0	Pre-Requisites:	Nil
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05
Course Objectives:			
1	To understand the fundamentals of Vehicle dynamics and systems.		
2	To evaluate the influence vehicle subsystems on the vehicle behavior.		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Introduction: Earth and vehicle coordinate system. Longitudinal, lateral and vertical vehicle dynamics. Dynamic axle loads. Road loads - Aerodynamic forces and moments, viscosity effects, separation and its control; aerodynamic lift and its control, ground effect, styling for minimum drag. Rolling resistance, grade loads.	7
2	Performance Mode: Acceleration - Free body diagram of accelerating vehicle, maximum transferable tractive force, gradability, Deceleration - free body diagram of decelerating vehicle, maximum decelerating rates, stopping distance, maximum braking force. Vehicle performances.	7
3	Ride Mode: Degrees of freedom-single, two and multi degrees of freedom system, free, forced and damped vibration, model of an automobile, magnification factor, transmissibility, vibration absorbers, pitch and bounce motion, oscillation centers, active and semi active suspension, orthogonality of mode shapes, modal analysis. MATLAB based Quarter car modelling and simulation.	8
4	Vehicle stability & Springing system: Requirements, sprung mass and un-sprung mass, wheel hop, shimmy, wheel wobble, choice of suspension spring rate, calculation of effective spring rate. Tyres - mechanics, stability of vehicle on slope, on curve and banked road. Pacejka's tire model using MATLAB.	7
5	Handling Mode: Vehicle control-low speed cornering and static Steering Ackerman steering geometry, steady-state cornering -steering factors, vehicle control parameters (under steer, neutral steer and over steer), roll steer, compliance steer, ride steer, slip angle steer, steady state handling-lateral acceleration gain, characteristic speed, yaw velocity gain, critical speed, effect of braking on vehicle handling_ MATLAB based Assignment.	7
Total		36

Course Outcomes:	
After completion of the course, students will be able to:	
1	Understand the fundamentals of Vehicle dynamics and describe various terminologies.
2	Explain various performance mode characteristics of the vehicle.
3	Analyse ride mode oscillations and compare their characteristics.
4	Evaluate the suspension and stability systems concepts and design the same according to requirements.
5	Analyse the cornering characteristics of the vehicle and evaluate yaw stability.

Learning Resources:	
1	Thomas D Gillespie, “Fundamentals of Vehicle Dynamics”, SAE USA 1992.
2	Rao S.S “Mechanical Vibrations”, Pearson Education Publication, 2009.
3	Babu A.K., “Automobile Mechanics”, Khanna Publishers, New Delhi.
4	Cole D.E., “Elementary Vehicle Dynamics”, Ann Arbor, Michigan, USA, 1972.
5	De. A., Vehicle Dynamics, Galgotia Publications Pvt. Ltd., 2010.
6	Pacejka H., Tire and Vehicle Dynamics, Elsevier, 2012.



MCKV INSTITUTE OF ENGINEERING

NAAC Accredited "A" Grade Autonomous Institute under UGC Act 1956
 Approved by AICTE & affiliated to Maulana Abul Kalam Azad University of Technology, West Bengal
 243 G.T. Road (N), Liluah, Howrah- 711204, West Bengal, India
 Ph: +91 33 26549315/17 Fax +91 33 26549318 Web: www.mckvie.edu.in/

Course Name:	Advanced Manufacturing Technology and Management		
Course Code:	PE-AUE701A	Category:	PE
Semester:	7	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Manufacturing Methods (PC-AUE 302) Machine Tools and Machining Technology (PC-AUE 402)
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05
Course Objectives:			
1	To understand the contemporary manufacturing philosophies related to automation and to interpret the application of computers in various aspects of Manufacturing including design, planning, layout, cost, material handling system, and control.		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Introduction: Automation in Manufacturing, Basic Elements of an Automated system, Levels of Automation, Types of production facilities, Manufacturing models and Metrics, Mathematical models of Production Performance, Basics of Manufacturing Control, Introduction to CAD/CAM/CIM, Lean Production and Just-In-Time Production.	6
2	Fundamentals of Line Balancing: Fundamentals of Manual Assembly Lines, Analysis of Single-Model Assembly Lines- Cycle Time and Workload Analysis, Repositioning Losses, Line Balancing Algorithms, Fundamentals of Automated Production Lines- Work part Transport, storage Buffers, Transfer lines	6
3	Cellular Manufacturing: Group Technology (GT), Part Families, Parts Classification and coding, Production flow Analysis, Cellular Manufacturing, Composite part concept, Machine cell design and layout, Quantitative analysis in Cellular Manufacturing, Rank Order Clustering Method, Hollier Method.	5
4	Flexible Manufacturing Systems (FMS): Types of Flexibility, FMS Components, FMS Application & Benefits, FMS Planning and Control, Quantitative analysis in FMS. Automated Guided Vehicle System (AGVS), AGVS Application	6
5	Computer Integrated Manufacturing: Components of CIM, Functional classification of CIM database, Communication network in CIM, Database management systems, benefits, Industry 4.0, functions, applications, benefits	4

6	Production Planning and Control: Process planning, Computer Aided Process Planning (CAPP), Logical steps in Computer Aided Process Planning, Concurrent Engineering, Aggregate Production Planning, and the Master Production Schedule, Material Requirement Planning, Capacity Planning, Control Systems-Shop Floor Control Inventory Control, Brief on Manufacturing Resource Planning-II (MRP-II) & Enterprise Resource Planning (ERP).	9
Total		36

Course Outcomes:

After completion of the course, students will be able to:

1	Explain the basic elements of an automated manufacturing system and recent manufacturing philosophies.
2	Analyze the manufacturing flow lines to reduce downtime and enhance productivity.
3	Explain the basic concept, elements, and applications of cellular manufacturing, flexible manufacturing systems and computer-integrated manufacturing.
4	Determine the planning and control of production cost, inventory, material requirement, shop floor, and enterprise resource planning.
5	Solve problems in different aspects related to advanced manufacturing technologies and production planning and control.

Learning Resources:

1	Groover M.P. “Automation, Production Systems and Computer Integrated Manufacturing”, Pearson Education; 4 th edition, 2016.
2	Radhakrishnan P, Subrahmanyam S., and Raju V., “CAD/CAM/CIM”, 2 nd Edition, New Age International (P) Ltd, New Delhi, 2009.
3	Koren Y, “Computer Control of Manufacturing System”, 1 st edition , McGraw Hill Education, 2017.
4	Groover M.P. and Zimmers E. Jr., “CAD/CAM”, Prentice Hall of India Pvt. Ltd., New Delhi-1, 2008.



MCKV INSTITUTE OF ENGINEERING

NAAC Accredited "A" Grade Autonomous Institute under UGC Act 1956
Approved by AICTE & affiliated to Maulana Abul Kalam Azad University of Technology, West Bengal

243 G.T. Road (N), Liluah, Howrah- 711204, West Bengal, India

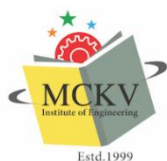
Ph: +91 33 26549315/17 Fax +91 33 26549318 Web: www.mckvie.edu.in/

Course Name:	Advanced Automotive Materials		
Course Code:	PE-AUE701B	Category:	Professional Elective Courses
Semester:	Seventh	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Engineering Materials
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05
Course Objectives:			
1	To present a problem oriented in depth knowledge of Automobile materials and manufacturing		
2	To address the underlying concepts and methods behind Automobile materials and manufacturing		
3	To develop an overall sound knowledge of metallurgical and materials Engineering.		
4	Evaluate and arrive at material properties for automotive components and select appropriate materials		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Demand of advanced automotive materials for automotive components, Different advanced materials used for automotive components, Functionality considerations of automotive parts, Factors influencing selection of advanced materials for automotive components. Factors influencing material change in future, emission control and fuel systems. Further directions in advanced automotive materials: environmental viewpoint, safety viewpoint and improving crashworthiness, Advanced materials and technologies for reducing noise, vibration and harshness (NVH) in automobiles.	7
2	Advanced metallic materials and vehicle light weighting, Lightweight materials and implications on vehicle design, Micro alloyed, High strength low alloy steel (HSLA), Advanced High Strength Steels (AHSS), Ultra high strength Steels (UHSS), TRIP steels, TWIP steel, Interstitial free (IF) steels, Smart materials, Shape memory alloy, Magnesium alloys, Aluminum alloys, Advanced materials for high temperature service, Ti and Zr alloys, Ni and Co based super alloys, Formability of steel sheets and tailor welded blanks for automotive application, Aluminium sheet for automotive applications, Strengthening mechanisms and their need in automotive environment, Elastomers and engineering thermoplastic and thermosets usage based on the functionality requirement, Advanced Ceramic materials: need for advanced ceramics.	7

3	Introduction to functional materials and nanotechnology, Nanostructured steel for automotive body structures, Functionally gradient material (FGM), Carbon based nanomaterial, Nanomaterial fabrication, Application of nano materials in automotive component.	7
4	Composites and hybrid composite, Nanocomposites, Characteristics of fibers, matrices, interface bonding, adhesives, Carbon fiber composites, Polymer-matrix composites, Carbon-matrix composite, Metal-matrix composite, Ceramic-matrix composite, Advanced composite processing techniques: Filament winding, pultrusion, pulforming, thermoforming, injection, injection molding, liquid molding, blow molding, Carbon based nanocomposites.	7
5	Selection of advanced materials for automotive components like cylinder block, head & liner, piston & piston rings, gudgeon pin, connecting rod, bearings, crankshaft, flywheel, camshaft, valves, valves seats, springs, gear train, chain & belt drives, suspension systems, cabin interiors. Need of advanced automotive materials characterization and available techniques: Optical Microscopy, Electron Microscopy, X-ray diffraction method, Surface Analysis, Spectroscopy, Dynamic mechanical analysis, Thermal analysis.	8
Total		36
Course Outcomes:		
After completion of the course, students will be able to:		
1	Make themselves familiar with advanced automotive materials	
2	Acquire knowledge on different class of materials and their selection criterion	
3	Evaluate and match materials and manufacturing processes	
4	Evaluate the cause for failure of the components due to material or manufacturing process and recommend the appropriate remedy to avoid the failure	

Learning Resources:	
1	W. E. Smith, Structure & Properties of Engineering Alloys, McGraw Hill, 1993
2	K. K. Chawla, Composite Materials, 2nd Edition, Springer - Verlag 2001
3	W. D. Callister, Material Science and Engineering: an Introduction, Wiley, 2002.
4	G. Dieter, Mechanical Metallurgy, Mc-Graw Hill, 1996.
5	P.R Khangaonkar, An Introduction to Materials Characterization, Penram International, 2010.
6	J. Rowe, Advanced Materials in Automotive Engineering, Woodhead Publisher, 1st Edition, 2012.
7	EverJ Barbero, Introduction to composite materials design, Taylor&Francis, 1999.



MCKV INSTITUTE OF ENGINEERING

NAAC Accredited "A" Grade Autonomous Institute under UGC Act 1956
 Approved by AICTE & affiliated to Maulana Abul Kalam Azad University of Technology, West Bengal
243 G.T. Road (N), Liluah, Howrah- 711204, West Bengal, India
 Ph: +91 33 26549315/17 Fax +91 33 26549318 Web: www.mckvie.edu.in/

Course Name:	Finite Element Method & its Applications		
Course Code:	PE-AUE701C	Category:	Professional Elective Courses
Semester:	Seventh	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Strength of Materials
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05
Course Objectives:			
1	To learn the mathematical background of finite element methods.		
2	To learn to model complex geometry problems and solution techniques.		
3	To understand, characteristics and selection of different finite elements used in finite element methods.		
4	To apply of the FEM technique to engineering problems and validate the solutions through simulation software for real time applications.		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Historical Background, Mathematical modeling of field problems in Engineering, Governing equations, Basic concepts of the Finite Element Method (FEM), General procedure for finite element analysis, Need of studying FEM, Different FEM methods: Direct stiffness method, Weighted residual methods and Variational method.	6
2	Formulation of the stiffness matrix: the direct approach, 1-D structural problems, Finite element analysis of Spring and Bar, Elements' stiffness matrices in local coordinates and global coordinates, Finite element analysis of truss structures: one dimensional and two dimensional truss.	8
3	Formulation of the stiffness matrix: the Galerkin's approach, Analysis of a simple beam with FEM. Development of beam equations, Flexure element stiffness matrix, Analysis of simply supported beam – Analysis of propped cantilevers, Fixed beams, and Continuous beams for various loadings, Derivation of shape functions, Isoparametric formulation, Development of the linear-strain triangle equations. Sub-parametric, Super parametric elements, Serendipity elements.	8
4	Formulation of the stiffness matrix: variational approach, Potential energy approach to derive spring, bar and beam element equations, Analysis of a uniform shaft subjected to torsion, Formulation of axisymmetric problems.	8

5	Application of FEM method to solve engineering problems, FEM applications in solid mechanics, Development of the plane stress and plane strain stiffness equations, Finite element solution of a plane stress problem, FEM applications in heat transfer for 1D and 2D analysis, Steady state heat transfer analysis: 1-D heat conduction of slab 1D fin elements, 2D heat conduction, Analysis of thin plates, and analysis of a uniform shaft subjected to torsion problems, FEM applications in fluid mechanics, Introduction of use of FEM software such as ANSYS, ABAQUS.	6
Total		36
Course Outcomes:		
After completion of the course, students will be able to:		
1	Apply direct stiffness, Rayleigh-Ritz, Galerkin method to solve engineering problems.	
2	Identify the application and characteristics of FEA elements such as bars, beams, plane and isoperimetric elements, and 2-D element.	
3	Apply suitable boundary conditions to global equations, and reduce it to a solvable form.	
4	Solve real life problems using finite element analysis.	

Learning Resources:	
1	V. David. Hutton, Fundamentals of finite elements analysis, Tata McGraw-Hill education (P) Ltd, 1st edition, 2010.
2	Logan D L., A first course in the Finite Element Method, Thomson-Engineering, 4th edition, 1994.
3	J. N. Reddy, An introduction to Finite Element Method, Tata McGraw hill education (P) Ltd, New Delhi, India. 3rd edition, 2010.
4	Desai, Y.M., Eldho. TI, Shah, A.H, Finite Element Methods with Application in Engineering, Pearson, 1st Edition, 2011.
5	S.S. Bhavikatti, Finite Element Analysis, New age international, 2nd Edition, 2010.
6	Chandrupatla, T.R., Belegunde, A.D, Introduction to Finite Elements in Engineering, PHI, 3rd Edition, 2010.
7	Seshu P., Text Book of Finite Element Analysis, PHI Learning Pvt. Ltd., 2003.
8	Ever J. Barbero, Finite Element Analysis of Composite Materials using Abaqus, CRC Press, 2015.



MCKV INSTITUTE OF ENGINEERING

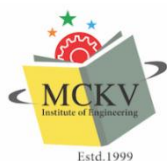
NAAC Accredited "A" Grade Autonomous Institute under UGC Act 1956
 Approved by AICTE & affiliated to Maulana Abul Kalam Azad University of Technology, West Bengal
243 G.T. Road (N), Liluah, Howrah- 711204, West Bengal, India
 Ph: +91 33 26549315/17 Fax +91 33 26549318 Web: www.mckvie.edu.in/

Course Name:	Automotive Component & System Design		
Course Code:	PE-AUE702A	Category:	Professional Elective Courses
Semester:	Seventh	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Machine Design
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05
Course Objectives:			
1	Provide students with the fundamental knowledge in the field of automotive design.		
2	An appreciation of the relationships between component level design and overall automotive system design and performance.		
3	Develop analytical abilities to give solutions to automotive design problems.		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Design of principal parts of I.C. Engines: Cylinder and cylinder liner-Material Selection, Design of Piston, piston rings and piston pin or gudgeon pin, Material Selection. Design considerations, Design calculations of Connecting rod with small and big end bearing-forces acting on it.	8
2	Design of Crank, crankshaft and crank pin. Design of inlet and outlet valves, Design of valve springs, rocker arm, tappet, Cam, camshaft, Design procedure for flywheel.	7
3	Design of Clutches and Gear Boxes: single plate, multiple plates, centrifugal clutch, lining material, lever design, sliding mesh, constant mesh, synchromesh gear box, gear ratio and gear shifting lever.	7
4	Design of Drive train: Design of propeller shaft and U-joints, criteria, failure theories-joint design, Design of Final drive and differential.	5
5	Brakes and Suspension: Force analysis, design of drum and disc brakes, internal expanding shoe brake, friction lining material, leaf spring, coil spring, materials, suspension system and linkages, independent suspension.	5
6	Advanced automotive Body Structures: Emphasis is on body concept for design. Material selection and manufacturing constraints.	4
Total		36

Course Outcomes:	
After completion of the course, students will be able to:	
1	Analyze various forces coming on the vehicle and deduce the power of the engine.
2	Evaluate the working stresses and design of an automotive component such as cylinder, piston, connecting rod, cam etc.
3	Estimate the working stresses and design of an automotive component such as brakes, spring, propeller shaft, etc.
4	Analyze various forces coming on the Clutch and Gear box and design the components.

Learning Resources:	
1	Shigley J.E., Mechanical Engineering Design, McGraw Hill, 2003.
2	Giles J G "Engine Design", Illiffee Books Ltd., London, 1968
3	John Fenton "Handbook of Automotive Body and Systems Design", John Wiley & Sons, 2013.
4	Bhandari V.B., Design of Machine Elements , McGraw Hill Pub.
5	Sadhu Singh, Machine Design, Khanna Book Publishing House, 2016.
6	Khurmi R S, Gupta J K "A Text Book of Machine Design", Eurasia Publishing house., New Delhi, 2010.



MCKV INSTITUTE OF ENGINEERING

NAAC Accredited "A" Grade Autonomous Institute under UGC Act 1956
Approved by AICTE & affiliated to Maulana Abul Kalam Azad University of Technology, West Bengal

243 G.T. Road (N), Liluah, Howrah- 711204, West Bengal, India

Ph: +91 33 26549315/17 Fax +91 33 26549318 Web: www.mckvie.edu.in/

Course Name:	Automotive Pollution control & Alternate Fuels		
Course Code:	PE-AUE702B	Category:	Professional Elective Courses
Semester:	Seventh	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Nil
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05
Course Objectives:			
1	Create awareness about alternative fuels and their need.		
2	Student should be able to differentiate the conventional fuels and Alternative fuels.		
3	Student should get the knowledge about application of alternative fuels.		
4	Student should understand the Emission norms up to Bharat Stage VI.		
5	Student should get knowledge of emission from different automotive vehicles and its control.		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Conventional Fuels and Need for alternative fuels: Estimate of petroleum reserve and availability- comparative properties of fuels diesel and gasoline, quality rating of SI and CI engine fuels, fuel additives for SI and CI engines, thermodynamics of fuel combustion introduction to chemical thermodynamics, chemical reaction- fuels and combustion, enthalpy of formation and enthalpy of combustion, first law analysis of reacting systems, adiabatic flame temperature, need for alternative fuels, applications, types etc.	8
2	Alternative Fuels I– Gaseous Fuels and Biofuels: Introduction to CNG, LPG, ethanol, vegetable oils, bio-diesel, biogas, Hydrogen and HCNG, study of availability, manufacture, properties, storage, handling and dispensing, safety aspects, engine/ vehicle modifications required and effects of design parameters performance and durability	6
3	Alternative Fuels II- Synthetic Fuels: Introduction to Syngas, DME, P Series, GTL, BTL, study of production, advantages, disadvantages, need, types, properties, storage and handling, dispensing and safety, discussion on air and water vehicles.	6
4	Emission Control (SI Engine): Emission formation in S.I. engines - Hydrocarbons, carbon monoxide, oxides of nitrogen, poly-nuclear aromatic hydrocarbon, effects of design and operating variables on emission formation in spark ignition engines, controlling of pollutant formation in engines exhaust after treatment, charcoal canister control for evaporative emission control, emissions and drivability, positive crank case ventilation system for UBHC emission reduction.	6

5	Emission Measurement and Control (CI Engine): Chemical delay, intermediate compound formation, pollutant formation on incomplete combustion, effect of design and operating variables on pollutant formation, controlling of emissions, emissions and drivability, exhaust gas recirculation, exhaust after treatment– DOC, DPF, SCR and LNT measurement and test procedure (NDIR analyzers, FID, chemiluminescence NOx analyzer, oxygen analyzer, smoke measurement, constant volume sampling, particulate emission measurement, Orsat apparatus.).	6
6	Health effects of Emissions from Automobiles: Emission effects on health and environment. Emission inventory, ambient air quality monitoring, Emission Norms: As per Bharat Standard up to BS – VI.	4
Total		36

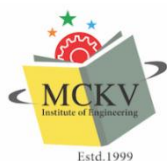
Course Outcomes:

After completion of the course, students will be able to:

1	Explain the importance of Alternative Fuels, its properties and their application.
2	Select an alternative fuel for specific application.
3	Classify and get knowledge of emission norms.
4	Identify the emission parameters from engine emission, its causes and remedies.

Learning Resources:

1	Thipse S.S., Alternative Fuels, Jaico Publications.
2	Pundir B.P., Engine Emission, Narosa Publication.
3	Ganesan V., Internal Combustion Engines, Tata McGraw Hill.
4	Crouse W.M. and Anglin A.L, Automotive Emission Control, McGraw Hill.
5	Thipse S.S., IC Engines, Jaico Publications.
6	Springer G.S. and Patterson D.J., Engine Emissions, Pollutant Formation, Plenum Press.
7	ARAI Vehicle Emission Test Manual.



MCKV INSTITUTE OF ENGINEERING

NAAC Accredited "A" Grade Autonomous Institute under UGC Act 1956
 Approved by AICTE & affiliated to Maulana Abul Kalam Azad University of Technology, West Bengal
243 G.T. Road (N), Liluah, Howrah- 711204, West Bengal, India
 Ph: +91 33 26549315/17 Fax +91 33 26549318 Web: www.mckvie.edu.in/

Course Name:	Two and Three Wheelers		
Course Code:	PE-AUE702C	Category:	Professional Elective Courses
Semester:	Seventh	Credit:	03
L-T-P:	3-0-0	Pre-Requisites:	Automotive Engines, Automotive Body & Chassis Engineering
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05
Course Objectives:			
1	The course is designed to understand different types of two and three wheelers' types, construction and working. Students will also be able to learn about different functions of two and three wheelers.		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Introduction: TWO WHEELERS: Types- standard class, touring class, sports bike, cruisers class, off road class. Main frame, types- tubular steel frame, diamond, cradle, backbone, underbone, trellis frame. THREE WHEELERS: Types- passenger carrier, goods carrier and its features. Main components of three-wheeler.	6
2	Power Unit for Two & Three Wheelers: Two stroke SI engine components, principle of operation, Types of two stroke engine: based on scavenging method- crank case, separately scavenged engine, based on scavenging process- cross flow, loop flow (MAN, Schnuerle, Curtis type), uni-flow (opposed piston, poppet valve, sleeve valve), based on overall port timing- symmetrical and unsymmetrical diagrams. Scavenging –efficiency, pumps. Reed valve, Rotary disc valve engine, fuel system, lubrication system, air and water cooling system, magneto coil and battery coil spark ignition system, electronic ignition system, variable timing ignition system (VTI), starting systems-kick starter, electric starting.	8
3	Transmission & Suspension Systems for Two & Three-Wheeler: Multiple and centrifugal clutches, gear box, gear shift mechanism, Final drive- shaft drive and chain drive. Front suspension-Head stock mounted forks, leading link, Trailing link, Telescopic forks, rear suspension-conventional dual spring /damper type, pro-link progressive type, Pro arm type, torsion bar, leaf spring.	8
4	Sub-Systems for Two & Three Wheelers: Panel meters and controls on handle bar, Controls on foot. Electrical systems lights- replaceable bulb, sealed beam, tail light and stop light, turn signal lights, horn, fuses. Drum brakes- Brake lever, wheel brake mechanism-single leading shoe, double	8

	leading shoe brakes, disc brakes- master cylinder calliper assemble, disc, brake fluid, front and rear brake links layouts. Wheels & Tyres - spoked wheel, cast wheel, disc wheel, tyre construction, tubed and tubeless tyre, composite wheel construction, split rim wheel.	
5	Service And Maintenance: Servicing and maintenance of carburetor, spark plug cleaning, ignition timing adjustment, gear box, steering head, shaft and chain drive, suspension, brake servicing and adjustment. Periodic maintenance schedules.	6

Total	36
--------------	-----------

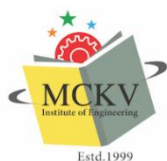
Course Outcomes:

After completion of the course, students will be able to:

1	Describe different types of two wheelers and three wheelers.
2	Explain the working, components and accessories of two and three-wheeler power units.
3	Describe the transmission and suspension system of two and three wheelers.
4	Describe meters, controls, brakes, wheels and tyres for two and three wheelers.
5	Explain the procedure for servicing and maintenance of different components used in two and three wheelers.

Learning Resources:

1	Irving P.E, “Motor cycle engineering”, Temple Press Book, London, 1992.
2	Giri N.K, “Automobile Mechanics”, Khanna Publishers, New Delhi, 2006.
3	“The cycle Motor Manual”, Temple Press Ltd, 1990.
4	“Encyclopedia of Motor Cycling 20 Volumes”, Marshall Cavensih, New York and London, 1989.
5	Raymond Board, Lambretta, “A Practical Guide to Maintenance and Repair”, 1987.
6	Ganesan V, “Internal Combustion Engines” Tata McGraw Hill, New Delhi, 2003.



MCKV INSTITUTE OF ENGINEERING

NAAC Accredited "A" Grade Autonomous Institute under UGC Act 1956
 Approved by AICTE & affiliated to Maulana Abul Kalam Azad University of Technology, West Bengal
243 G.T. Road (N), Liluah, Howrah- 711204, West Bengal, India
 Ph: +91 33 26549315/17 Fax +91 33 26549318 Web: www.mckvie.edu.in/

Course Name:	Quality Control & Reliability Engineering		
Course Code:	OE-ME701C	Category:	Open Elective Courses
Semester:	Seventh	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Nil
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05
Course Objectives:			
1	To apply the knowledge of descriptive statistics.		
2	To understand and apply different types of sampling methods used in control engineering.		
3	To develop a basic understanding concepts of reliability engineering.		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Basics of quality: Quality objectives, Quality control, Quality Assurance, Quality costs, Quality loss function, Statistical tolerance, quality assurance statistical tools used in quality in SQC, Quality planning, Organization for quality. Bureau of Indian standards, ISO 9000: 2008-quality circles KAIZEN-TQM concepts-Quality audit.	5
2	Statistical Process Control: Variation in processes, Factors, Process capability, Analysis of process capability, control charts, variables, Attributes, Establishing and interpreting control charts, X,R, chart for variables, defects, P chart, C-chart and U chart-Con-troll charts for defective quality rating.	7
3	Acceptance Sampling: Lot-by-lot sampling, types probability of acceptance in single double, multiple sampling techniques- O.C. curves, procedure's Risk and consumers Risk AQL, LTPD, AOQL concepts standard sampling plans for AQL and LTPD- uses of standard sampling plans.	5
4	Life Testing-Reliability-Systems Approach: Life testing-objectives classification-failure characteristics-failure data analysis-mean time to failure-maintainability and availability-reliability-system reliability series and parallel systems-systems reliability in terms of probability of failure-MTBF-Acceptance sampling based on reliability test OC curves	8
5	Quality and Reliability: Reliability improvement- techniques, use of Pareto analysis- Design for reliability, Redundancy, standby redundancy, failsafe systems- optimization in reliability, product design, product analysis, product development product cycle.	7
6	Quality function deployment: FMEA, Quality circles, ISO 9000 series and 14000 series, 3 Sigma and 6 Sigma concepts.	4
Total		36

Course Outcomes:	
After completion of the course, students will be able to:	
1	Express the knowledge about various concepts of descriptive statistics.
2	Apply and use different tools and techniques used in quality control engineering.
3	Develop a basic understanding of concepts of reliability engineering along with use of statistical and design model.
4	Understand and apply different types of sampling methods used in control engineering.

Learning Resources:	
1	Betsterfield D.H., Quality Control, Revised Edition, Prentice Hall Pub, 1993
2	Sharma S.C. &Poonia M.P., Total Quality Management, Khanna Publishing House, New Delhi 2016.
3	Bank J., The Essence of Total Quality Management, Prentice Hall of India Pvt. Ltd., New Delhi, 1995.
4	Samson D., Manufacturing & Operations Strategy, Prentice Hall, New York, 1991.
5	Bagchi T.P., ISO9000 Concepts Methods and Implementation, Wheeler Publisher Allahabad, 1994
6	Ross P.J., Taguchi Techniques for Quality Engineering, McGraw Hill, New York, 1998.



MCKV INSTITUTE OF ENGINEERING

NAAC Accredited "A" Grade Autonomous Institute under UGC Act 1956
Approved by AICTE & affiliated to Maulana Abul Kalam Azad University of Technology, West Bengal

243 G.T. Road (N), Liluah, Howrah- 711204, West Bengal, India

Ph: +91 33 26549315/17 Fax +91 33 26549318 Web: www.mckvie.edu.in/

Course Name:	Mechatronic Systems		
Course Code:	OE-ME701B	Category:	Open Elective Courses
Semester:	7	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Mathematics III(BS-M303)
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05
Course Objectives:			
1	To impart knowledge on the mechatronic systems with their components including sensors, transducers, actuators, microprocessors, control systems etc.		
2	To explain concepts of smart materials and micro-mechatronic systems and their applications in various fields		
3	To enable students to model and solve problems related to mechatronic systems		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Introduction: Definition of Mechanical Systems, Philosophy, and approach; Systems and Design: Mechatronic approach, Integrated Product Design, Modeling, Analysis and Simulation, Man Machine Interface.	4
2	Sensors and transducers: classification, Development in Transducer technology, Opto-electronic shaft encoders, CD Sensors, Vision systems.	6
3	Drives and Actuators: Hydraulic and Pneumatic drives, Electrical Actuators such as servo motor and Stepper motor, Drive circuits, open and closed loop control.	8
4	Embedded Systems: Hardware Structure, Software Design and Communication, Programmable Logic Devices, Automatic Control, and Real-Time Control Systems.	8
5	Smart materials: Shape Memory Alloy, Piezoelectric and Magnetostrictive Actuators: Materials, Static and dynamic characteristics, illustrative examples for positioning, vibration isolation.	5
6	Micro-mechatronic systems: Micro-sensors, Micro-actuators; Micro-fabrication techniques LIGA Process: Lithography, etching, Micro-joining, etc. Application examples; Case studies Examples of Mechatronic Systems from Robotics Manufacturing, Machine Diagnostics, Road vehicles, and Medical Technology.	5
Total		36

Course Outcomes:	
After completion of the course, students will be able to:	
1	Explain the basic concepts of mechatronic systems for an engineering application
2	Identify sensors, transducers, and actuators to monitor and control the behavior of the process or product.
3	Apply the concept of PLC and control systems for engineering applications.
4	Interpret smart material for mechatronic systems and their application
5	Demonstrate micro-mechatronic systems with fabrication techniques and their application in real life.

Learning Resources:	
1	Shetty D. & Richard A. K., Mechatronics System Design, 2 nd Edition, Cengage, 2012
2	Bolton, W., Mechatronics: A Multidisciplinary Approach, 4 th edition, Pearson Education, 2016
3	Rajput, R.K., A Textbook of Mechatronics, S. Chand & Company Private Limited , 2007
4	Bolton, W., Mechatronics: Electronic Control Systems in Mechanical and Electrical Engineering, 6 th Edition, Pearson Education, 2019
5	Alciatore D.G., & Hstand, M.B. Introduction to Mechatronics and Measurement systems, 4th Edition, McGraw Hill, 2006.
7	M.D. Singh and J.G. Joshi, Mechatronics, Prentice Hall of India, 2006.
8	K.K. Appu Kuttan, Introduction to Mechatronics, Oxford University Press, New Delhi, 2007.
9	F.H. Raven, Automatic Control Engineering, McGraw Hill India, 2013.
10	K. Ogata, Modern Control Engineering, Prentice Hall, 2010.



MCKV INSTITUTE OF ENGINEERING

NAAC Accredited "A" Grade Autonomous Institute under UGC Act 1956
Approved by AICTE & affiliated to Maulana Abul Kalam Azad University of Technology, West Bengal

243 G.T. Road (N), Liluah, Howrah- 711204, West Bengal, India

Ph: +91 33 26549315/17 Fax +91 33 26549318 Web: www.mckvie.edu.in/

Course Name:	Machine Learning		
Course Code:	OE-IT701C	Category:	Open Elective Courses
Semester:	Seventh	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Mathematics III (BS-M303)
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05
Course Objectives:			
1	To introduce students to the basic concepts and techniques of Machine Learning.		
2	To have a thorough understanding of the Supervised and Unsupervised learning techniques.		
3	To understand graphical models of machine learning algorithms.		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Introduction: Learning– Types of Machine Learning– Supervised Learning– The Brain and the Neuron– Design a Learning System– Perspectives and Issues in Machine Learning – Concept Learning Task–Concept Learning as Search– Finding a Maximally Specific Hypothesis – Version Spaces and the Candidate Elimination Algorithm –Linear Discriminants– Perceptron– Linear Separability– Linear Regression.	7
2	Linear Models: Multi-layer Perceptron– Going Forwards– Going Backwards: Back Propagation Error– Multilayer Perceptron in Practice–Examples of using the MLP– Overview– Deriving Back Propagation –Radial Basis Functions and Splines– Concepts– RBF Network– Curse of Dimensionality– Interpolations and Basis Functions– Support Vector Machines.	7
3	Tree and Probabilistic Models: Learning with Trees– Decision Trees–Constructing Decision Trees– Classification and Regression Trees–Ensemble Learning– Boosting– Bagging– Different ways to Combine Classifiers – Probability and Learning– Data into Probabilities– Basic Statistics– Gaussian Mixture Models– Nearest Neighbour Methods–Unsupervised Learning– K means Algorithms– Vector Quantization–Self Organizing Feature Map	7

4	Dimensionality Reduction and Evolutionary Models: Dimensionality Reduction– Linear Discriminant Analysis– Principal Component Analysis– Factor Analysis– Independent Component Analysis– Locally Linear Embedding– Isomap– Least Squares Optimization– Evolutionary Learning– Genetic algorithms– Genetic Offspring: Genetic Operators–Using Genetic Algorithms– Reinforcement Learning– Overview–Getting Lost Example– Markov Decision Process.	8
5	Graphical Models: Markov Chain Monte Carlo Methods– Sampling–Proposal Distribution– Markov Chain Monte Carlo– Graphical Models–Bayesian Networks– Markov Random Fields– Hidden Markov Models–Tracking Methods.	7
Total		36
Course Outcomes:		
After completion of the course, students will be able to:		
1	Distinguish between, supervised, unsupervised and semi-supervised learning.	
2	Apply the appropriate machine learning strategy for any given problem.	
3	Suggest supervised, unsupervised or semi-supervised learning algorithms for any given problem.	
4	Design systems that uses the appropriate graph models of machine learning.	
5	Modify existing machine learning algorithms to improve classification efficiency.	

Learning Resources:	
1	Marsland S., “Machine Learning– An Algorithmic Perspective”, 2e, Chapman and Hall/CRC Machine Learning and Pattern Recognition Series, 2014.
2	Mitchell T.M., “Machine Learning”, First Edition, McGraw Hill Education, 2013.
3	Flach P., “Machine Learning: The Art and Science of Algorithms that Make Sense of Data”, First Edition, Cambridge University Press, 2012.
4	Bell J., “Machine learning- Hands on for Developers and Technical Professionals”, First Edition, Wiley, 2014.
5	Alpaydin E., “Introduction to Machine Learning”, 3rd Edition, Adaptive Computation and Machine Learning Series, MIT Press, 2014.
6	Dr. Jeeva Jose, “Introduction to Machine Learning”, Khanna Publishing House, 2019.



MCKV INSTITUTE OF ENGINEERING

NAAC Accredited "A" Grade Autonomous Institute under UGC Act 1956
 Approved by AICTE & affiliated to Maulana Abul Kalam Azad University of Technology, West Bengal
 243 G.T. Road (N), Liluah, Howrah- 711204, West Bengal, India
 Ph: +91 33 26549315/17 Fax +91 33 26549318 Web: www.mckvie.edu.in/

Course Name:	Cloud Computing		
Course Code:	OE-CS701C	Category:	Open Elective Courses
Semester:	Seventh	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Computer Architecture, Operating System, Networking
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05
Course Objectives:			
1	This course gives students an insight into the basics of cloud computing along with virtualization.		
2	It will provide the students basic understanding of cloud security and privacy issues.		
3	Students will be able to use different cloud services for different purposes.		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	<u>Definition of Cloud Computing and its Basics:</u> Defining a Cloud, Cloud Types – NIST model, Cloud Cube model, Deployment models (Public, Private, Hybrid and Community Clouds), Service model: - Infrastructure as a Service, Platform as a Service, Software as a Service with examples of services/ service providers. Cloud Reference model, Characteristics of Cloud Computing, Benefits and advantages of Cloud Computing, A brief introduction on Composability, Infrastructure, Platforms, Virtual Appliances, Communication Protocols, Applications, Connecting to the Cloud by Clients, IaaS –concept, Workload, partitioning of virtual private server instances, Pods, aggregations. SaaS - Basic concept and characteristics, Open SaaS and SOA, examples of SaaS platform Identity as a Service (IDaaS) Compliance as a Service (CaaS)	9
2	<u>Use of Platforms in Cloud Computing:</u> Concepts of Abstraction and Virtualization technologies: Types of virtualization (access, application, CPU, storage), Load Balancing and Virtualization: Basic Concepts, Network resources for load balancing, Advanced load balancing (including Application Delivery Controller and Application Delivery Network), Mention of The Google Cloud as an example of use of load balancing. Hypervisors: Virtual machine technology and types, VMware vSphere Machine Imaging. Porting of applications in the Cloud: The simple Cloud API and AppZero Virtual Application appliance, Concepts of Platform as a Service, Definition of services, Distinction between SaaS and PaaS (knowledge of Salesforce.com and Force.com), Application	12

	development Use of PaaS Application frameworks, Discussion of Google Applications Portfolio – Indexed search, Dark Web, Aggregation and disintermediation, Productivity applications and service, Adwords, Google Analytics, Google Translate, a brief discussion on Google Toolkit (including introduction of Google APIs in brief), major features of Google App Engine service., Windows Azure platform: Microsoft’s approach, architecture, and main elements, overview of Windows Azure AppFabric, Content Delivery Network, SQL Azure, and Windows Live services,	
3	<u>Cloud Infrastructure:</u> Cloud Management: An overview of the features of network management systems and a brief introduction of related products from large cloud vendors, Monitoring of an entire cloud computing deployment stack – an overview with mention of some products, Lifecycle management of cloud services (six stages of lifecycle). Concepts of Cloud Security: Cloud security concerns, Security boundary, Security service boundary Overview of security mapping Security of data: Brokered cloud storage access, Storage location and tenancy, encryption, and auditing and compliance Identity management (awareness of Identity protocol standards).	7
4	<u>Concepts of Services and Applications :</u> Service Oriented Architecture: Basic concepts of message-based transactions, Protocol stack for an SOA architecture, Event-driven SOA, Enterprise Service Bus, Service catalogs, Applications in the Cloud: Concepts of cloud transactions, functionality mapping, Application attributes, Cloud service attributes, System abstraction and Cloud Bursting, Applications and Cloud APIs Cloud-based Storage: Cloud storage definition – Manned and Unmanned Webmail Services: Cloud mail services including Google Gmail, Mail2Web, Windows Live Hotmail, Yahoo mail, concepts of Syndication services	8
Total		36
Course Outcomes:		
After completion of the course, students will be able to:		
1	Describe the fundamental concept of cloud computing and its characteristics, benefits and limitations.	
2	Explain different types of cloud models, architecture and infrastructure of cloud computing and its examples.	
3	Explain abstraction and different types of virtualization, load balancing technology and their role in the cloud computing model.	
4	Explain the security, privacy and cloud management of cloud computing.	
5	Use various cloud services in different applications.	

Learning Resources:	
1	Cloud Computing Bible by Barrie Sosinsky, Wiley India Pvt. Ltd, 2013
2	Mastering Cloud Computing by Rajkumar Buyya, Christian Vecchiola, S. ThamaraiSelvi, McGraw Hill Education (India) Private Limited, 2013
3	Cloud computing: A practical approach, Anthony T. Velte, Tata Mcgraw-Hill
4	Cloud Computing, Miller, Pearson
5	Building applications in cloud: Concept, Patterns and Projects, Moyer, Pearson
6	Cloud Computing – Second Edition by Dr. Kumar Saurabh, Wiley India



MCKV INSTITUTE OF ENGINEERING

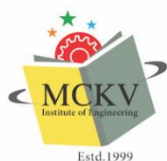
NAAC Accredited "A" Grade Autonomous Institute under UGC Act 1956
 Approved by AICTE & affiliated to Maulana Abul Kalam Azad University of Technology, West Bengal
243 G.T. Road (N), Liluah, Howrah- 711204, West Bengal, India
 Ph: +91 33 26549315/17 Fax +91 33 26549318 Web: www.mckvie.edu.in/

Course Name:	Values and Ethics		
Course Code:	HM-HU704	Category:	Humanities and Social Sciences including Management Courses
Semester:	Seventh	Credit:	2
L-T-P:	2-0-0	Pre-Requisites:	Nil
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05
Course Objectives:			
1	To help the students appreciate the essential complementarity between 'VALUES' and 'SKILLS' to ensure sustained happiness and prosperity, which are the core aspirations of all human beings.		
2	To facilitate the development of a Holistic perspective among students towards life and profession as well as towards happiness and prosperity based on a correct understanding of the Human reality and the rest of Existence. Such a holistic perspective forms the basis of Universal Human Values and movement towards value-based living in a natural way.		
3	To highlight plausible implications of such a Holistic understanding in terms of ethical human conduct, trustful and mutually fulfilling human behaviour and mutually enriching interaction with Nature.		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Concept of Human Values, Ethics and Profession; Engineering as a profession	2
2	Societal values: justice, democracy, secularism, rule of law; Maslow's Hierarchy and Integrated personality; Value Crisis in contemporary society	5
3	Codes of professional ethics. Whistle blowing and beyond. Case studies.	3
4	Human Operator in Engineering projects and industries. Problems of man machine interaction.	2
5	Rapid Technological growth and depletion of resources. Reports of the Club of Rome. Limits of growth; sustainable development.	4
6	Environmental degradation and pollution. Eco-friendly Technologies; Appropriate Technology Movement of Schumacher: later developments.	3
7	Emotional Intelligence – Salovey – Mayer Model; Uses of Ethical Theories – Deontology- Types of Inquiry –Kohlberg's Theory – Gilligan's Argument – Heinz's Dilemma -Intellectual Property Rights.	5
Total		24

Course Outcomes:	
After completion of the course, students will be able to:	
1	The students are able to see that verification on the basis of natural acceptance and experiential validation through living is the only way to verify right or wrong, and referring to any external source like text or instrument or any other person cannot enable them to verify with authenticity; it will only develop assumptions
2	The students are able to grasp the right utilization of their knowledge in their streams of Technology/Engineering/ Management to ensure mutually enriching and recyclable productions systems
3	The students are able to see that lack of right understanding leading to lack of relationship is the major cause of problems in their family and not the lack of physical facilities in most of the cases, while they have given higher priority to earning of physical facilities in their life ignoring relationships and not being aware that right understanding is the most important requirement for any human being

Learning Resources:	
1	Stephen H Unger, Controlling Technology: Ethics and the Responsible Engineers, John Wiley & Sons, New York 1994 (2nd Ed)
2	Deborah Johnson, Ethical Issues in Engineering, Prentice Hall, Englewood Cliffs, New Jersey 1991
3	A N Tripathi, Human values in the Engineering Profession, Monograph published by IIM, Calcutta ,1996
4	Samita Manna and Suparna Chakraborti: Values and Ethics in Profession
5	S.K. Sarangi, Values and Ethics in Profession



MCKV INSTITUTE OF ENGINEERING

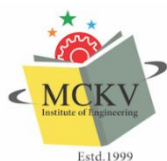
NAAC Accredited "A" Grade Autonomous Institute under UGC Act 1956
 Approved by AICTE & affiliated to Maulana Abul Kalam Azad University of Technology, West Bengal
 243 G.T. Road (N), Liluah, Howrah- 711204, West Bengal, India
 Ph: +91 33 26549315/17 Fax +91 33 26549318 Web: www.mckvie.edu.in/

Course Name:	Automobile Engineering Lab V (Automotive Electrical & Electronics Lab)		
Course Code:	PC-AUE791	Category:	Professional Core Courses
Semester:	Seventh	Credit:	1.5
L-T-P:	0-0-3	Pre-Requisites:	Nil
Full Marks:	100		
Examination Scheme:	Semester Examination: 60	Continuous Assessment: 35	Attendance: 05
Course Objectives:			
1	To provide knowledge about application of electrical and electronics in automobile engineering.		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Battery testing	3
2	Alternator testing and Starter motor testing	3
3	Diagnosis of ignition system.	3
4	Diagnosis of automotive electrical wiring.	3
5	Fault finding of relay & fuses in car using Off Board Diagnostics Systems (OBDS).	3
6	Study the Characteristics of Rectifier and IC Timer.	3
7	Study of Logic Gates and combinational circuits-Adder Circuitry.	3
8	Study of Sequential circuits- Flip-Flops.	3
9	Study of D/A Converter and A/D Converter	3
10	Assembly Language Programming exercise using Microcontroller	3
11	Interfacing of Stepper Motor Using Microcontroller	3
12	Study of Data Acquisition System using Microcontroller	3
Total		36

Course Outcomes:	
After completion of the course, students will be able to:	
1	Identify and explain the testing procedure of battery, alternator and starter motor
2	Explain different kinds of automotive wiring and fault diagnosis capabilities of relay and fuses
3	To understand the basics of instrumentation, measurement, data acquisition, interpretation and analysis
4	To learn rectifiers, filters, A/D and D/A convertors.

Learning Resources:	
1	Lab manual as per above experiments to be prepared by the institution



MCKV INSTITUTE OF ENGINEERING

NAAC Accredited "A" Grade Autonomous Institute under UGC Act 1956
 Approved by AICTE & affiliated to Maulana Abul Kalam Azad University of Technology, West Bengal
243 G.T. Road (N), Liluah, Howrah- 711204, West Bengal, India
 Ph: +91 33 26549315/17 Fax +91 33 26549318 Web: www.mckvie.edu.in/

Course Name:	Project-III		
Course Code:	PW-AUE781	Category:	Project/ Internships/ Sessional
Semester:	Seventh	Credit:	3
L-T-P:	0-0-6	Pre-Requisites:	Nil
Full Marks:	100		
Examination Scheme:	Semester Examination: 100		
Course Objectives:			
1	To develop the ability to identify, formulate and analyze engineering problems through literature survey, recent trends in industries and by applying the knowledge of science and engineering fundamentals.		
2	To train students in preparing project reports, to face reviews and viva voce examination		

Course Contents:	
Description of Topic	Contact Hrs.
It is intended to start the project work and carry out both design and fabrication of a mechanical/automobile device whose working can be demonstrated. The design and formulation of the problem is expected to be completed. The students in a group of 4 to 6 works on a topic.	72
Total	72
Course Outcomes:	
After completion of the course, students will be able to:	
1	carry out some project works based on some design or fabrication or experimental problems in a group building up team spirit and would get sufficient exposure for the way to proceed to solve a practical or design problem



MCKV INSTITUTE OF ENGINEERING

NAAC Accredited "A" Grade Autonomous Institute under UGC Act 1956
 Approved by AICTE & affiliated to Maulana Abul Kalam Azad University of Technology, West Bengal
243 G.T. Road (N), Liluah, Howrah- 711204, West Bengal, India
 Ph: +91 33 26549315/17 Fax +91 33 26549318 Web: www.mckvie.edu.in/

Course Name:	Constitution of India		
Course Code:	MC772	Category:	Mandatory Courses
Semester:	Seventh	Credit:	0
L-T-P:	2-0-0	Pre-Requisites:	Nil
Full Marks:	100		
Examination Scheme:	Semester Examination: 100		

Course Objectives:

1	To realise the significance of constitution of India to students from all walks of life and help them to understand the basic concepts of Indian constitution.
2	To identify the importance of fundamental rights as well as fundamental duties.
3	To understand the functioning of Union, State and Local Governments in Indian federal system.
4	To learn procedure and effects of emergency, composition and activities of election commission and amendment procedure.

Course Contents:

Module No.	Description of Topic	Contact Hrs.
1	Introduction to Constitution: Meaning and importance of the Constitution, salient features of Indian Constitution. Preamble of the Constitution. Fundamental rights- meaning and limitations. Directive principles of state policy and Fundamental duties -their enforcement and their relevance.	6
2	Union Government: Union Executive- President, Vice-president, Prime Minister, Council of Ministers. Union Legislature- Parliament and Parliamentary proceedings. Union Judiciary-Supreme Court of India – composition and powers and functions.	6
3	State and Local Governments: State Executive- Governor, Chief Minister, Council of Ministers. State Legislature-State Legislative Assembly and State Legislative Council. State Judiciary-High court. Local Government-Panchayat raj system with special reference to 73rd and Urban Local Self Govt. with special reference to 74th Amendment.	6
4	Election provisions, Emergency provisions, Amendment of the constitution: Election Commission of India-composition, powers and functions and electoral process. Types of emergency-grounds, procedure, duration and effects. Amendment of the constitution- meaning, procedure and limitations.	6
Total		24L

Course Outcomes:

After completion of the course, students will be able to:

1	Understand and explain the significance of Indian Constitution as the fundamental law of the land.
2	Exercise his fundamental rights in proper sense at the same time identifies his responsibilities in national building.
3	Analyse the Indian political system, the powers and functions of the Union, State and Local Governments in detail
4	Understand Electoral Process, Emergency provisions and Amendment procedure.

Learning Resources:

1	M.V. Pylee, Introduction to the Constitution of India, 4th Edition, Vikas publication, 2005.
2	Durga Das, Basu DD, Introduction to the constitution of India, Student Edition, 19th edition, Prentice-Hall EEE, 2008.
3	Merunandan, Multiple Choice Questions on Constitution of India, 2nd Edition, Meraga publication, 2007.